Quirk plane search with MadGraph and Pythia: efficiencies, uncertainties, and Figures 2–5

Comparison with Knapen, Lou, Papucci, and Setford, arXiv:1708.02243

Abstract

This note reports the MadGraph5_aMC@NLO and Pythia 8 samples of the quirk plane search. The electroweak sample is Drell–Yan production of an 800 GeV vector-like lepton. The colored sample is QCD production of an 1800 GeV vector-like quark. Both use a string scale of 4 keV. Every cut is counted sequentially. Each efficiency is given with its binomial standard error and a Wilson 95% confidence interval, and is compared with Table I of the paper. The generator settings are listed in full. Figures 2–5 of the paper are reproduced for these benchmarks: the internal angular momentum and separation, the rotation of the quirk plane, one event projected onto its reconstructed plane, and the distributions of the thickness Δs and the width Δw.

1 Generator settings

The hard process and the parton shower come from MadGraph5_aMC@NLO 3.8.0 and Pythia 8.317. The quirk oscillation is not part of Pythia. After the shower, each matched event is propagated with the straight-string equation of motion in a 2 T solenoid, at $\Lambda = 4$ keV, and passed through the eight-layer barrel and the plane finder.

1.1 Collider and run card

Both processes use the same collider settings.

Setting	Value
Beams	proton–proton
Beam energy	6500 GeV each, so $\sqrt{s} = 13$ TeV
PDF	NNPDF2.3 LO, LHAPDF set `nn23lo1`
Random seed	170802243
Events requested	2000 unweighted events
Cores	16
Extra-jet matching	MLM, `ickkw = 1`
Matching scale Q_{cut}	`xqcut = 30` GeV
Pythia jet-matching scale	`JetMatching:qCut = 45` GeV
Maximum extra jets	`JetMatching:nJetMax = 1`
Light flavours in the match	`JetMatching:nQmatch = 4`
Jet algorithm	k_T, exclusive, cone radius 1, `doShowerKt = off`
Jet p_T cut in the matrix element	`ptj = 10` GeV
Scale and PDF systematics	off (`use_syst = False`)
MadSpin	off
Analysis	off

1.2 Electroweak process

The electroweak quirk is the charge−1 vector-like lepton `taup` in the model `VLLS_NLO`. Its PDG code is 17 and it is a color singlet. The mass parameter is `MTAUP` = 800 GeV. The width is set to zero by `epstaupeL` = `epstaupmuL` = `epstauptaL` = 0. The generated processes are

$$\begin{aligned} &pp \to \tau'\bar{\tau}', && \texttt{QED=2}, \\ &pp \to \tau'\bar{\tau}'\, j, && \texttt{QED=2, QCD=1}. \end{aligned}$$

Hadronization is left at the Pythia default. The lepton is a color singlet, so there is no QCD string between the pair. MadGraph reports a parton-level cross section of $3.814 \times 10^{-4} \pm 1.901 \times 10^{-6}$ pb for 2000 events. MLM matching keeps 1040 events. The matched cross section is $1.983 \times 10^{-4} \pm 4.374 \times 10^{-6}$ pb. The efficiencies below use these 1040 matched events as N_{gen}.

1.3 Colored process

The colored quirk is the charge +2/3 vector-like quark `tp` in the model `VLQ_UFO`. Its PDG code is 6000006 and it is a color triplet. The mass parameter is `MTP` = 1800 GeV and the width is `WTP` = 0. The generated processes are

$$\begin{aligned} &pp \to t'\bar{t}', \\ &pp \to t'\bar{t}'\, j. \end{aligned}$$

Pythia hadronization is switched off with `HadronLevel:all = off`. A stable colored pair does not leave enough energy for a QCD string to fragment, and with hadronization left on the shower never writes an event. The dark string is the straight-string propagator in this package, not the Pythia string. The parton shower and the MLM match are still performed. MadGraph reports a parton-level cross section of $7.566 \times 10^{-4} \pm 3.700 \times 10^{-6}$ pb for 2000 events. MLM matching keeps 855 events. The matched cross section is $3.234 \times 10^{-4} \pm 8.518 \times 10^{-6}$ pb. The efficiencies below use these 855 matched events as N_{gen}.

These two models are vector-like fermions used as proxies for the quirk. Their cross sections are the MadGraph results for those models. They are not the quirk cross sections used for the discovery contours in the paper. The JSON files also contain a field `cross_section_pb` from an earlier toy estimate. That field is not the MadGraph cross section and is not used in this note.

1.4 Quirk propagation and the detector

Each matched event supplies the two heavy-fermion three-momenta. The pair is started at the origin with a separation of 10^{-3} mm along the relative velocity. The straight-string force and the Lorentz force are integrated in a 2 T field along z, until both quirks leave the outer barrel or 25 ns have passed. Hits are measured relative to that vertex. The 45 mm beam spot is not sampled again on top of the MadGraph event. Pixel hits are moved to the pixel centre: 50 µm in $r\phi$, 0.25 mm in z on the innermost layer and 0.40 mm on the other pixel layers. SCT hits are smeared by 34 µm in $r\phi$, twice the 17 µm intrinsic resolution. The SCT stereo z uncertainty is about 1 mm. It is not applied as a shift of the hit, because it is larger than the 0.1 mm thickness cut. Neighboring hits are merged. No pileup and no photon conversions are overlaid on these samples. The colored charged-hadron fraction of 30% is not applied inside the efficiency. It is used only when a yield is quoted.

2 Efficiency definitions

An event is counted at a cut only if it has passed every earlier cut. The three efficiencies in the paper are ratios of those counts,

$$\varepsilon_{\text{trig}} = \frac{N_{\text{trig}}}{N_{\text{gen}}}, \qquad N_{\text{trig}} = \#\{p_T > 200\,\text{GeV}\}, \tag{1}$$

$$\varepsilon_{\text{fid}} = \frac{N_{\text{fid}}}{N_{\text{trig}}}, \tag{2}$$

$$\varepsilon_{\text{reco}} = \frac{N_{\text{reco}}}{N_{\text{fid}}}, \tag{3}$$

$$\varepsilon = \varepsilon_{\text{trig}}\,\varepsilon_{\text{fid}}\,\varepsilon_{\text{reco}} = \frac{N_{\text{reco}}}{N_{\text{gen}}}. \tag{4}$$

Here N_{gen} is the number of events kept by MLM matching, 1040 for the electroweak sample and 855 for the colored sample. The fiducial count N_{fid} requires both quirks to intersect all eight barrel layers, with every hit time below 25 ns. The reconstruction count N_{reco} is the number of events that pass the final hit-pattern cut. Because the flow is sequential, that count already requires the trigger, the fiducial requirement, the seed windows, the factor-of-three growth test, $\Delta w < 1\,\text{cm}$, and $\Delta s < 0.01\,\text{cm}$.

The individual cuts, in order, are:

1. **Generated.** Every matched event.
2. **Trigger.** Pair transverse momentum above 200 GeV.
3. **Fiducial.** Both quirks hit all eight layers within 25 ns.
4. **Layer-8 seed.** A hit pair on layer 8 with $\Delta\phi < 0.1$ and $\Delta z < 2\,\text{cm}$.
5. **Layer-7 seed.** A hit pair on layer 7 in the same window.
6. **Seed prefilter.** The two pairs agree within 0.5 rad and 80 mm. This removes combinations that cannot satisfy the 1 cm width cut.
7. **Seed thickness.** The four seed hits have $\Delta s < 0.05\,\text{cm}$.
8. **Seed width.** The four seed hits have $\Delta w < 1\,\text{cm}$.
9. **Forward direction.** Every seed hit satisfies $\boldsymbol{n}_3 \cdot \boldsymbol{x} > 0$.
10. **Inward walk.** From layer 6 down to layer 1, a hit is added only when both Δs and Δw stay within a factor of three of their current values, and the finished plane has $\Delta w < 1\,\text{cm}$.
11. **Final thickness.** The finished plane has $\Delta s < 0.01\,\text{cm}$.
12. **Hit pattern.** Seven layers contain two hits and the remaining layer contains at least one.

3 Wilson intervals

Each efficiency is a binomial proportion k/n, with estimate $\hat{p} = k/n$. The standard error is

$$\sigma = \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n}}. \tag{5}$$

The 95% interval is the Wilson score interval. With $z = 1.960$,

$$\begin{aligned} \hat{p}_{\mathrm{W}} &= \left(\hat{p} + \frac{z^2}{2n}\right) \frac{1}{1 + z^2/n}, \\ \delta &= \frac{z}{1 + z^2/n} \sqrt{\frac{\hat{p}\,(1-\hat{p})}{n} + \frac{z^2}{4n^2}}, \end{aligned} \tag{6}$$

and the interval is $\hat{p}_{\mathrm{W}} \pm \delta$, clipped to $[0, 1]$. For a conditional cut, k is the number of events that pass the cut and n is the number that reached it. The paper does not quote uncertainties on Table I.

4 Electroweak sample

Mass 800 GeV, string scale 4 keV, $N_{\mathrm{gen}} = 1040$ matched events.

Table 1: Sequential cut flow for the electroweak MadGraph sample. The efficiency is k/n, where n is the number of events that reached the cut.

Cut	N_{pass}	N_{in}	Efficiency	Std. error	Wilson 95% CI
Generated	1040	1040	1.000	0.000	[1.000, 1.000]
Trigger	182	1040	0.175	0.012	[0.153, 0.199]
Fiducial	58	182	0.319	0.035	[0.255, 0.390]
Layer-8 seed	53	58	0.914	0.037	[0.814, 0.963]
Layer-7 seed	51	53	0.962	0.026	[0.872, 0.990]
Seed prefilter	40	51	0.784	0.058	[0.654, 0.875]
Seed Δs	40	40	1.000	0.000	[0.912, 1.000]
Seed Δw	40	40	1.000	0.000	[0.912, 1.000]
Forward	40	40	1.000	0.000	[0.912, 1.000]
Inward walk	40	40	1.000	0.000	[0.912, 1.000]
Final Δs	39	40	0.975	0.025	[0.871, 0.996]
Hit pattern	22	39	0.564	0.079	[0.410, 0.707]

Table 2: Electroweak efficiencies as defined in Eqs. (1)–(4).

Efficiency	Count	Estimate	Std. error	Wilson 95% CI
$\varepsilon_{\mathrm{trig}}$	182/1040	0.175	0.012	[0.153, 0.199]
$\varepsilon_{\mathrm{fid}}$	58/182	0.319	0.035	[0.255, 0.390]
$\varepsilon_{\mathrm{reco}}$	22/58	0.379	0.064	[0.266, 0.508]
ε	22/1040	0.021	0.004	[0.014, 0.032]

The last row of Table 1 is not $\varepsilon_{\mathrm{reco}}$. It is the hit-pattern efficiency among events that already passed the final Δs cut. The paper's reconstruction efficiency uses the fiducial sample as the denominator.

5 Colored sample

Mass 1800 GeV, string scale 4 keV, $N_{\mathrm{gen}} = 855$ matched events.

Table 3: Sequential cut flow for the colored MadGraph sample.

Cut	N_{pass}	N_{in}	Efficiency	Std. error	Wilson 95% CI
Generated	855	855	1.000	0.000	[1.000, 1.000]
Trigger	208	855	0.243	0.015	[0.216, 0.273]
Fiducial	54	208	0.260	0.030	[0.205, 0.323]
Layer-8 seed	50	54	0.926	0.036	[0.824, 0.971]
Layer-7 seed	48	50	0.960	0.028	[0.865, 0.989]
Seed prefilter	43	48	0.896	0.044	[0.778, 0.955]
Seed Δs	43	43	1.000	0.000	[0.918, 1.000]
Seed Δw	43	43	1.000	0.000	[0.918, 1.000]
Forward	43	43	1.000	0.000	[0.918, 1.000]
Inward walk	43	43	1.000	0.000	[0.918, 1.000]
Final Δs	43	43	1.000	0.000	[0.918, 1.000]
Hit pattern	29	43	0.674	0.071	[0.525, 0.795]

Table 4: Colored efficiencies as defined in Eqs. (1)–(4).

Efficiency	Count	Estimate	Std. error	Wilson 95% CI
$\varepsilon_{\text{trig}}$	208/855	0.243	0.015	[0.216, 0.273]
ε_{fid}	54/208	0.260	0.030	[0.205, 0.323]
$\varepsilon_{\text{reco}}$	29/54	0.537	0.068	[0.406, 0.663]
ε	29/855	0.034	0.006	[0.024, 0.048]

6 Comparison with Table I

Table I of arXiv:1708.02243 gives $\varepsilon_{\text{trig}} = 0.10$ and $\varepsilon_{\text{fid}} = 0.28$ for the 800 GeV electroweak benchmark, independent of Λ, and $\varepsilon_{\text{reco}} = 0.72$ at $\Lambda = 4$ keV. For the 1800 GeV colored benchmark it gives $\varepsilon_{\text{trig}} = 0.24$ and $\varepsilon_{\text{fid}} = 0.28$. It does not list a colored reconstruction efficiency at 4 keV. The neighboring colored entries are $\varepsilon_{\text{reco}} = 0.59$ at 3 keV and $\varepsilon_{\text{reco}} = 0.74$ at 5 keV.

Table 5: Table I compared with the MadGraph samples. A Table I value is marked outside when it does not lie in the Wilson 95% interval.

Sample	Efficiency	Table I	This work	Std. error	Wilson 95% CI	Interval
Electroweak	$\varepsilon_{\text{trig}}$	0.10	0.175	0.012	[0.153, 0.199]	outside
Electroweak	ε_{fid}	0.28	0.319	0.035	[0.255, 0.390]	inside
Electroweak	$\varepsilon_{\text{reco}}$	0.72	0.379	0.064	[0.266, 0.508]	outside
Colored	$\varepsilon_{\text{trig}}$	0.24	0.243	0.015	[0.216, 0.273]	inside
Colored	ε_{fid}	0.28	0.260	0.030	[0.205, 0.323]	inside
Colored	$\varepsilon_{\text{reco}}$	0.59 at 3 keV; 0.74 at 5 keV	0.537	0.068	[0.406, 0.663]	not tabulated at 4 keV

The colored trigger efficiency and both fiducial efficiencies agree with Table I: each Table I value lies inside the Wilson interval. The electroweak trigger efficiency is higher than 0.10, and 0.10 lies outside the interval. The vector-like lepton model, with one extra jet and MLM matching, is not the same matrix element as the quirk model used for Table I, so the momentum distribution is not expected to match that entry exactly. The electroweak reconstruction efficiency is 22/58. The Table I value 0.72 lies outside its Wilson interval. The colored reconstruction efficiency at 4 keV is not in Table I. Its central value, 0.537, lies below the paper's 3 keV entry of 0.59, and 0.59 lies inside the Wilson interval. The 5 keV entry 0.74 lies outside that interval. The fiducial samples that enter $\varepsilon_{\text{reco}}$ contain 58 and 54 events, so this efficiency is the least precise of the three.

At $300\,\mathrm{fb}^{-1}$, the central matched cross section and the central total efficiency give about 1.26 electroweak events. For the colored sample the same product, multiplied by the 30% charged fraction, gives about 0.99 events. These yields use the vector-like-fermion cross sections above. They are not the quirk yields of the paper.

7 Figures

The four figures below correspond to Figures 2–5 of the paper. Figures 2 and 3 integrate the straight-string equation at the two benchmark masses, $\Lambda = 4\,\mathrm{keV}$, in the 2 T field. A showered MadGraph event oscillates many times while its centre of mass crosses the tracker, so those two figures use one resolved trajectory at each benchmark, with the initial conditions stated in the captions. Figure 4 is reconstructed MadGraph event 192, which has pair $p_T = 328\,\mathrm{GeV}$ and passes the hit-pattern cut. Figure 5 uses every fiducial event in the two matched samples for which the plane finder returned a plane.

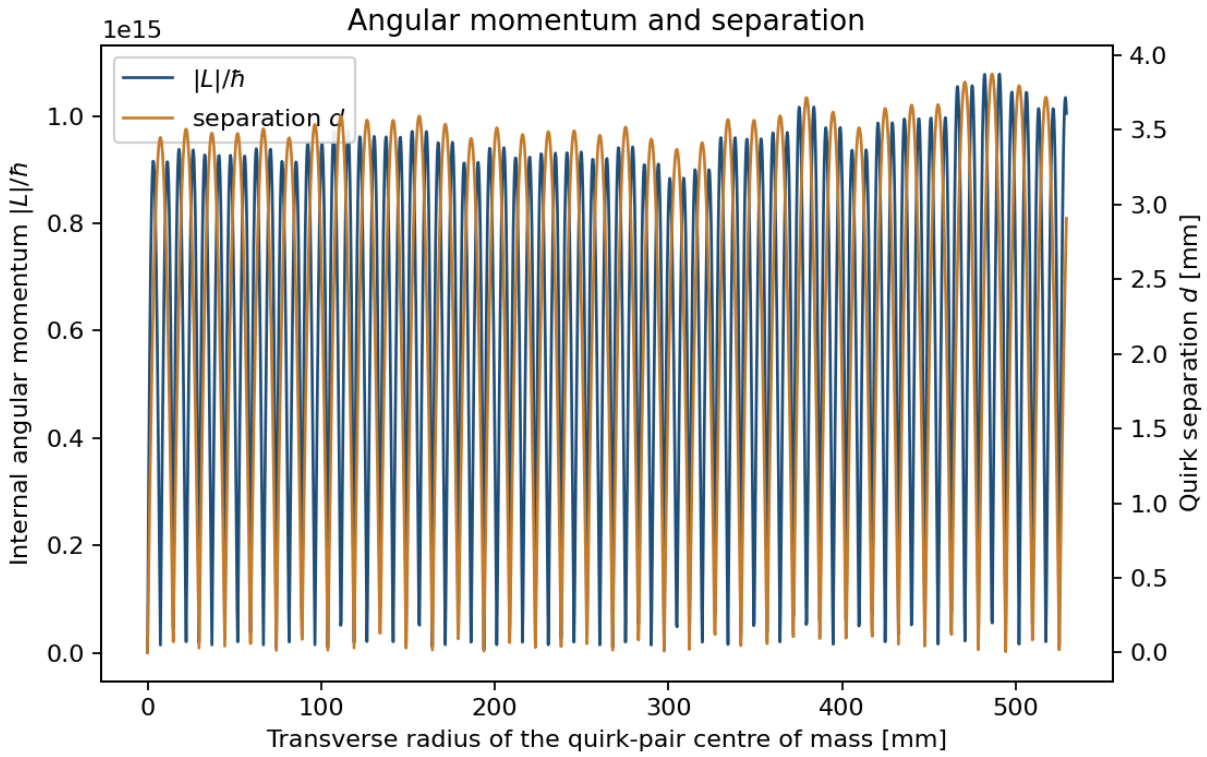


Figure 2: Internal angular momentum $|L|/\hbar$ and quirk separation d versus the transverse radius of the pair centre of mass. The quirks are produced back to back in their centre of mass with speed $v = 0.53$ along the axis $(0, 0.2, 1)$, and the pair is boosted by $\boldsymbol{\beta} = (0.75, 0.05, 0.20)$. The mass is 800 GeV and the string scale is 4 keV. On this trajectory $|L|/\hbar$ reaches 1.08×10^{15}, and d runs from 0.001 mm to 3.874 mm. This is the same pair of quantities as Figure 2 of the paper.

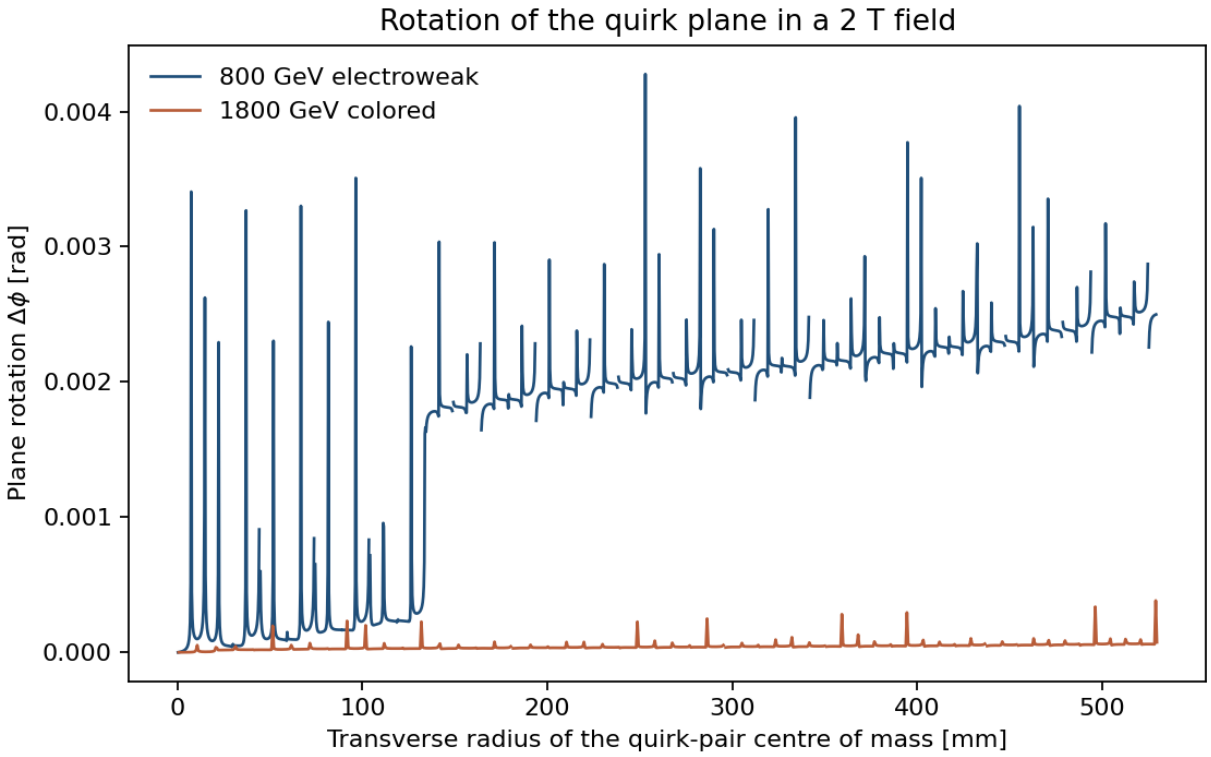


Figure 3: Angle $\Delta\phi$ between the instantaneous quirk plane and the plane at the start of the trajectory. The electroweak curve uses the same initial conditions as Figure 2 and reaches 4.28×10^{-3} rad. The colored curve has mass 1800 GeV, speed $v = 0.42$ along $(0.15, 1, 0.1)$, and boost $\boldsymbol{\beta} = (0.70, 0.10, 0.15)$. It reaches 3.83×10^{-4} rad. As in Figure 3 of the paper, the rotation remains small compared with the angular size of a tracker hit.

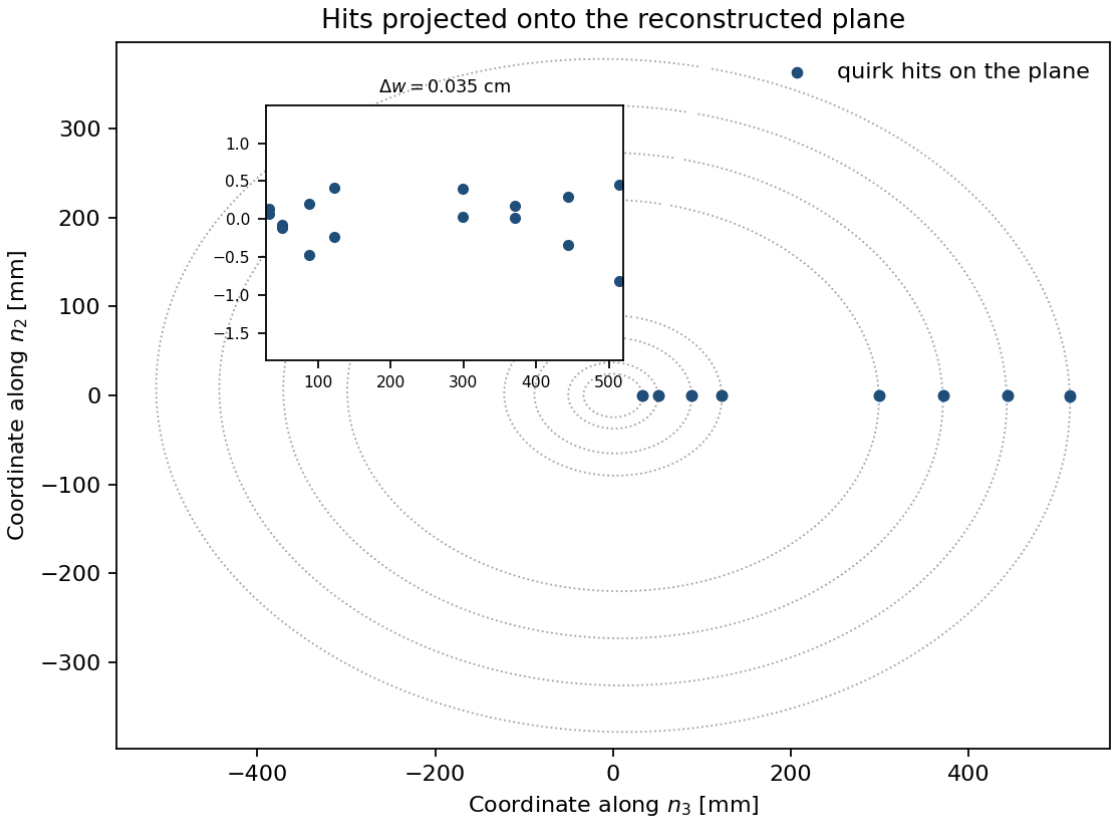


Figure 4: Hits of the electroweak MadGraph event projected onto the reconstructed $(\boldsymbol{n}_3, \boldsymbol{n}_2)$ plane. The dotted curves are the eight barrel layers. The fit gives $\Delta s = 0.0075\,\text{cm}$ and $\Delta w = 0.0345\,\text{cm}$, from 16 hits. The inset shows the strip, as in Figure 4 of the paper.

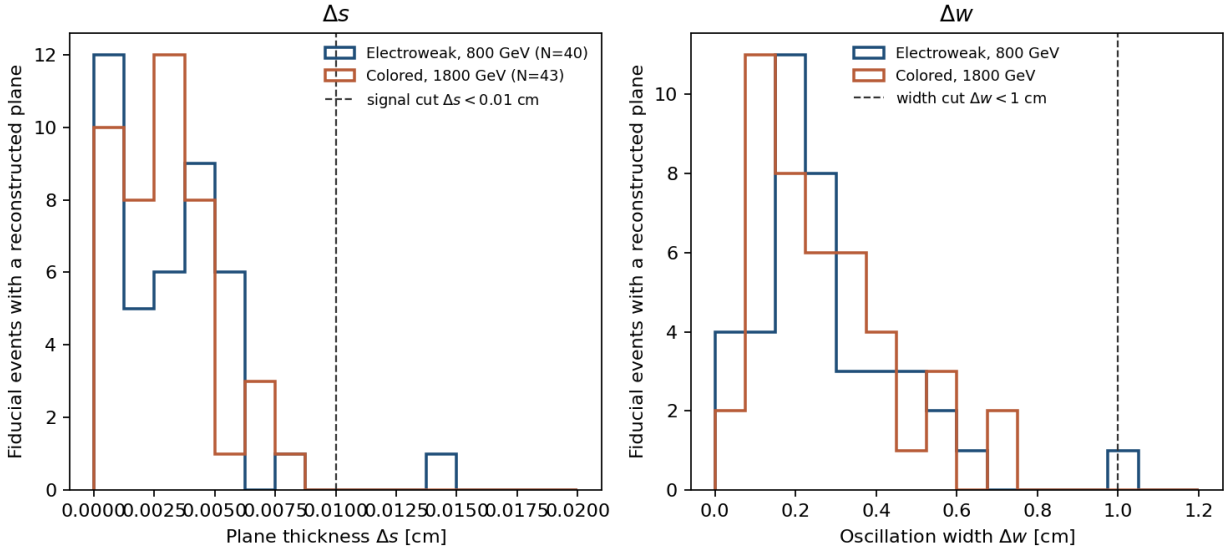


Figure 5: Thickness Δs and width Δw for fiducial MadGraph events in which the plane finder returned a plane, before the hit-pattern requirement. The dashed line on the left is the signal cut $\Delta s < 0.01\,\text{cm}$. The dashed line on the right is $\Delta w < 1\,\text{cm}$. There are 40 electroweak events and 43 colored events in the histograms. Figure 5 of the paper also shows the pileup and photon-conversion background, which was not generated in this run, and its colored curve is at 2 keV rather than 4 keV.